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Genome wide analysis for the identification and characterization of Superoxide-dismutase gene family in *Rosa chinensis* ascertains the role of salinity-responsive RcMSD1 protein and its interaction with peroxyl radical☆

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ABSTRACT

Abiotic stresses e.g., cold, heat, and salinity affect the quality and yield of *Rosa chinensis*, a well-known ornamental and medicinal plant. Previously, several attempts have been made to identify the genes that confer resistance to oxidative stressors. Superoxide-dismutase (SOD) is a crucial member of the class metalloenzyme that responds to abiotic stresses and protects plants by countering the reactive-oxygen species (ROS). In this genome-wide association study, the SOD gene family has been investigated in *R. chinensis*. Seven SOD genes, including three Cu/ZnSODs, two MnSODs, and two FeSODs were identified. Phylogenetic analysis revealed that RcSODs are divided into three clades; (i) CSDs (Cu/ZnSODs) (ii) FSDs (FeSODs) and MSDs (MnSODs). Most of the RcSODs exhibited different exons/introns distribution patterns. Motifs 3 and 5 are conserved in all *RcSODs* genes. The RNA-seq data analysis and qRT-PCR-based expression profiling indicated that RcSODs exhibit diverse responses under salt stress conditions. The *RcCSD1, RcCSD3,* and *RcFSD3* are significantly up-regulated under salt stress conditions in roots and down-regulated in leaves. This data provides valuable information for further application and function of SODs in *R. chinensis* under abiotic stresses.

Introduction

In external environment, both abiotic and biotic stresses are constantly impacting the plant's growth and development. Abiotic stresses including cold, salt, drought, and heat have significant implications on plant growth and development (Jia et al., 2020). During evolution, plants have developed several physiological and metabolic mechanisms to combat environmental constraints (Rasul et al., 2017; Nawaz et al., 2023). Abiotic stress-related metabolic activities resulted in the increasing production of ROS (Mhamdi and Van Breusegem, 2018). These ROS being highly reactive and toxic in nature disrupt the equilibrium of oxidative reaction resulting in oxidative stress (Oreiro et al., 2020; Shokri-Gharelo and Noparvar, 2018). ROS includes hydrogen peroxide (H_2O_2), singlet oxygens (1 O_2), peroxyl radicals (HOO-), and superoxide anion radicals (O^{2-}) (Karuppanapandian et al., 2011). The presence of an adequate amount of ROS is essential for plant growth and development and also contributes tolerance of plants to abiotic stresses (Huang et al., 2019). The progressive toxic level of ROS causes the oxidation of cell membranes, which in turn leads to irreparable dysfunction and cell death.

The toxicity of ROS in plants is counteracted by a sequence of antioxidant-resistant mechanisms. *In vivo*, a variety of antioxidant

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defense systems (both non-enzymatic and enzymatic) control the level of ROS in plants (Han et al., 2020). The antioxidant enzymatic response system consists of the catalase system (CAT), ascorbate-peroxidase (APX), superoxide dismutase (SOD), glutathione-peroxidase (GPX), glutathione-S transferase, reductase, and dehydro-ascorbate reductase (DHAR) (Finn et al., 2016). Among all of these, SOD was the first reported antioxidant enzyme to defend against oxidative stress (Geng et al., 2018). SODs catalyzed the dismutation of superoxide radicals to hydrogen peroxide, nontoxic water, and molecular oxygen (Lu et al., 2020).

The SOD genes can be divided into subfamilies based on the different metal cofactors: Mn-SODs, Fe-SODs, Cu/Zn-SODs, and Ni-SODs (Zelko et al., 2002). The Ni-SODs are present in Streptomyces (Ramana Gopavajhula et al., 2013). The SOD genes of different subfamilies are distributed into different cellular compartments. Mn-SODs are known to be found in the peroxisomes and mitochondria, FeSODs are primarily localized in the chloroplast, and Cu/Zn-SODs are mainly present in the cytoplasm (Wang et al., 2016). By decreasing oxidative stress, plants improve stress tolerance and enhance antioxidative defense capacity (Ashraf, 2009). Due to the crucial roles of SOD in the antioxidant system, several SOD genes have been cloned from different dicots and monocots (Feng et al., 2015). Overexpression of Cu/Zn-SOD genes is linked to better resistance to environmental stress in many plant types. For instance, Sunkar et al. (2006) found that activating two Cu/Zn-SOD genes post-transcriptionally in *Arabidopsis thaliana* was crucial for handling oxidative stress. Introducing a Cu/Zn-SOD gene (from peas) into tobacco plants enhanced their ability to withstand O3 exposure (Pitcher and Zilinskas, 1996). More recently, Madanala et al. (2011) observed that *Withania somnifera* plants with a stable chloroplastic Cu/Zn-SOD gene were better equipped to tolerate external substances like detergent and ethanol (Gill et al., 2015). It is reported that overexpression of SOD genes in plants conferred resistance to multiple environmental stresses like salinity, heat, drought, chilling and heavy metals (Negi et al., 2015). In tomatoes, the ectopic expression of *MnSOD* gives resistance to oxidative and salt stress (Guan et al., 2017). Some genes of SOD give tolerance to seagrasses under heat stress (Franssen et al., 2011). The first gene of SOD was cloned from maize (Sunkar et al., 2006). At present SOD gene family had been studied in different species like bananas (Feng et al., 2015), tomatoes (Yang et al., 2021), cotton (Wang et al., 2017), cucumber (Zhou et al., 2017), wheat (Jiang et al., 2019), foxtail millet (Wang et al., 2018), and grape (Han et al., 2020). Moreover, eight genes of the SOD family were also identified in *A. thaliana* (Haskirli et al., 2021).

R. chinensis species are the world's most commonly cultivated ornamental plants with high cultural, economic, and symbolic value (Jia et al., 2020). The pleasant floral fragrance, recurrent flowering trait, and diverse floral colors are due to the significant ornamental value of the rose plant (Tian et al., 2018). It occupies a major position in the floriculture industry and also accounts for auctions in the world flower market. Abiotic stresses are important limiting factors in increasing the production and growth of the rose. Salt stress influenced the successful cultivation of roses by affecting their productivity and growth (Asgari and Diyanat, 2021; Nadeem et al., 2023). Under the condition of heat stress, there is limited development and growth of roses where many plants enter dormancy and do not blossom (Ma et al., 2014). The market demand for roses is increasing and the cultivated quality of roses shows poor stress resistance. Therefore, exploring the salt stress-responsive mechanism in rose plants is imperative.

This study mainly related to the genome-wide discovery of the SODs gene family in *R. chinensis* (RcSOD). Combined structural and physicochemical characteristics, evolutionary relationship, 3D protein structure prediction, molecular docking analysis with peroxyl radical, and the SOD gene's behavior in response to salt stress.

Material and methods

Identification and characterization of SOD in R. chinensis

The amino acid sequences for the full-length SOD protein of *A. thaliana* were retrieved from the NCBI protein database ("https://www.ncbi.nlm.nih.gov"). The pre-identified sequences (AtFSD1, AtFSD3, AtFSD2, AtMSD, AtCSD1, AtCSD2, and AtCSD3) from *A. thaliana* database (TAIR: https://www.arabidopsis.org/) were used as queries in BLASTp with expect threshold of 0.05 and BLOSUM62 matrix was used to determine the SOD gene family members in *R. chinensis* and selected those genes with percentage identity greater than 70%. After removing redundant sequences the remaining hits were further submitted for SOD-related domain analysis through the NCBI conserved-domain database ("http://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi") & Pfam ("http://pfam.janelia.org/") (Jiang et al., 2019). Furthermore, the amino acid number, molecular weight, aliphatic index, grand average of hydropathicity (GRAVY) & theoretical isoelectric point (pI) of these genes were predicted by using ProtParam ("http://web.expasy.org/protparam/"). CELLO tool ("cello.life.nctu.edu.tw/") was employed to determine the subcellular location of all RcSODs (Verma et al., 2019).

Comparative phylogenetics, gene structure, and motif prediction analysis

SOD protein sequences of *Triticum aestivum, Gossypium hirsutum, Oryza sativa,* and *A. thaliana* were also retrieved from the public database NCBI. These protein sequences were aligned through ClustalW and adjusted the parameters (gap opening penalty 10 and gap extension penalty 0.02) (Thompson et al., 2003). MEGA 7.0 (Kumar et al., 2016) software was used to generate the maximum likelihood (ML) based phylogenetic tree to determine the phylogenetic relatedness of SOD genes using 1000 bootstrap value among these plant species (Verma et al., 2019). iTOL ("https://itol.embl.de/") was used for interactive editing of the phylogenetic tree. The conserved protein motifs were predicted using MEME ("http://meme-suite.org/tools/meme") with the default parameters, except for the number of motifs being set to 10 (Han et al., 2020). The intron, exon, and upstream region organization of *A. thaliana* and *R. chinensis* SOD genes were deciphered using the Gene Structure Display Server ("GSDS;http://gsds.cbi.pku.edu.cn/index.php") (Hu et al., 2015).

Gene duplication, syntenic analysis, and chromosomal localization

For chromosomal localization of genes, the information about the start and end positions of RcSODs with the corresponding chromosomes were retrieved from the NCBI-Gene database ("https://www.ncbi.nlm.nih.gov/gene"). DnaSP v.5 was used to calculate synonymous substitution (Ks) and non-synonymous substitution (Ka) values to check the selective pressure on duplicated gene pairs (Librado and Rozas, 2009). The duplication time of duplicated gene pairs was calculated by the formula: $T = \mathrm{Ks}/2\lambda \times 106$ Mya (Here the value of λ is $1.5 \times 10-8$ for dicots).

RNA-seq and real-time qRT-PCR based expression analysis of RcSODs

The expression analysis data were obtained from the NCBI's SRA under salt stress (Bio-project PRJNA428841) to examine the expression analysis of *R. chinensis* SOD genes during abiotic stress. From the NCBI database, the genome and gtf files were also downloaded. FASTQC (hosted on Galaxy server ("https://usegalaxy.org/")) (Zameer et al., 2022) was utilized to assess the quality of the reads. For read alignment against the reference genome, the Bowtie2 tool was employed. Gene expression levels were quantified by counting the mapped reads, and Cufflinks was used for this purpose, providing the results in the form of FPKM values. We identified the differentially expressed genes by performing a statistical analysis of the gene expression data. For genes to be considered upregulated, we applied a fold change cutoff of 1 or higher, indicating a significant increase in expression compared to the control group. Similarly, for downregulated genes, we used a fold change cutoff of 0 or lower, indicating a significant decrease in expression compared to the control group. These cutoff values ensure that the identified differentially expressed genes are biologically relevant and reflect meaningful changes in expression levels. Heatmap was constructed through TBtools ("https://github.com/CJ-Chen/TBtools") using the FPKM values. For real-time qRT-PCR, a salinity-sensitive cultivar, *R. chinensis* Jacq. was used. Stem cuttings collected from one-year-old plants, provided by a botanical garden of Government College University Faisalabad, Pakistan were used as the experimental material. Ten days after cultivating the cutting seedlings, NaCl treatment (200 mM) was applied for 2 and 4 h. For each treatment, three biological replicates were used. RNA was isolated using the Trizol reagent method and quantified using a Thermo Nanodrop 2000 (Thermo Fischer Scientific, Waltham, MA, USA). One ug of cDNA was synthesized from one ug of RNA with the First-Strand Synthesis kit and stored at −20 °C. The qRT-PCR was performed using an iTaq Universal SYBR Green Super-Mix (Bio-Rad Labs, Hercules, CA, USA) and a qRT PCR detection system (CFX96 TouchTM RT PCR Detection System). The NCBI Primer-BLAST algorithm ("https://www.ncbi.nlm.nih.gov/tools/primer-blast/"(accessed on 13 July 2022)) was used to confirm gene-specific primers designed using Oligo Calculator ("http://mcb.berkeley.edu/labs/krantz/tools/oligocalc.html/"(accessed on 13 August 2022)). The *GAPDH* (Locus tag: RchiOBHm_Chr4g0431271) was used as a housekeeping gene.

3D protein prediction and docking analysis

RcSOD proteins were used for a more thorough structural study, including molecular docking analysis and 3D protein structure prediction. To predict the 3D structures of the RcSODs proteins a web-based tool I-tasser ("https://zhanggroup.org/I-TASSER/") was used (Zhang, 2008). I-tasser is a hierarchical method for predicting protein structures and annotating functions based on those structures. First, it uses the multiple threading technique LOMETS to identify structural templates from the Protein Data Bank (PDB) ("https://www.rcsb.org/") (Berman et al., 2007), and then iterative template-based fragment assembly simulations are used to build full-length atomic models.

We chose a ligand peroxyl radical from PubChem ("https://pubchem.ncbi.nlm.nih.gov/") after the 3D structures of proteins were predicted. PyRx was then used to perform molecular docking on the predicted proteins (Dallakyan and Olson, 2015). The best docking result (results with the least RMSD values) was selected based on the binding affinity of the molecules, and the result was visualized by using UCSF Chimera (Pettersen et al., 2004).

Results

Identification of SOD in R. chinensis

By using the known protein sequences of *A. thaliana* as queries in the Blastp tool, we identified 7 potential *SOD* genes in *R. chinensis* after removing chimeric genes and remnant sequences (Jansen et al., 2005). Based on domain analysis from Pfam and NCBI-CDD (conserved domain database), these 7 candidate genes are divided into 3 groups (FeSODs, MnSODs, and Cu/ZnSODs) including 3 RcSOD identified within Copper-zinc, along with 2 RcSODs in Mn superoxide dismutase and 2 more in iron/manganese superoxide dismutase. Notably, the Sod_Cu domain (Pfam: PF00080) was conserved in all CSD. C terminus Fe_Mn-SOD domain (Pfam: PF02777) and N-terminus Fe_MnSOD-alpha hairpin domain (Pfam: PF00081) are present in 2 MSD and 1 FSD. Moreover, the Sod_Fe_N Superfamily domain exists in FSD 3 (Fig. 1). These genes were named *RcCSD1, RcCSD2, RcCSD3, RcMSD1, RcMSD2, RcFSD1* and *RcFSD3* based on their chromosomal location and domain analysis (Table 1). Moreover, the genomic position and sequence ID are also mentioned in (Table 1).

Subcellular localization and physiochemical properties of RcSOD genes

The subcellular location of RcSOD gene family members was predicted through CELLO 2.0 program. Among them, two MnSODs were predicted in the mitochondria, two Cu/ZnSODs were present in cytoplasmic and one Cu/ZnSODs and two members of FeSODs were found in the chloroplast. The RcSOD values of physiochemical properties are varied among all these parameters. In RcSOD the length of amino acid ranges from 152 to 320 aa, molecular weight varied from 15,251.98 to 36,136.50 Da and isoelectric point (pI) ranged from 4.97 to 7.90. The numeric value of GRAVY analyzes that only RcCSD3 shows a positive value or hydrophobic protein in nature and the remaining RcSOD are hydrophilic proteins (Table 1).

Phylogenetic analysis

To determine the evolutionary relationship of SODs in *R. chinensis, A. thaliana, O. sativa, G. Hirsutum,* and *T. aestivum,* a phylogenetic tree was generated (Fig. 2) using protein sequences of SODs from these plants. It revealed that the *RcSODs* are divided into three main clades, *MnSODs*, and *FeSODs* both were clustered together in a large clade and *Cu/ZnSODs* were present in another clade. The phylogenetic tree depicted that SOD

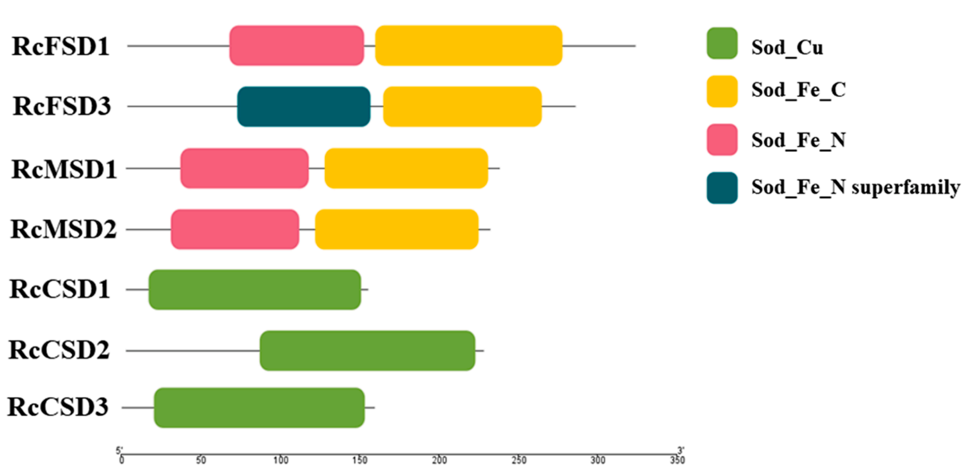

Fig. 1. Conserved domain analysis of RcSODs proteins in *R. chinensis* visualized through TBtool. Green color represented the Sod_Cu domain which was conserved in all CSDs while the yellow color represented the Sod_Fe_C domain which was conserved in all MSDs and FSDs.

Table 1
Physiochemical and biochemical properties of *SOD* genes in *R. chinensis*.

Gene name	Accession no	Protein Length(aa)	Chr no.	Exon	Start site	End site	Pfam	CDD	Isoelectric point (pI)	Molecular weight (Da)	Grand average of hydropathicity (GRAVY)	Location
RcFSD1	XP_024167896.1	320	1	9	61,520,605	61,522,930	Sod_Fe_N, Sod_Fe_C	PLN02685	4.97	36,136.5	−0.556	Chloroplast
RcFSD3	XP_024194410.1	282	4	9	66,854,733	66,858,235	Sod_Fe_N, Sod_Fe_C	PLN02622	7.17	32,631.24	−0.431	Chloroplast
RcMSD2	XP_024178663.1	229	1	6	60,888,991	60,891,686	Sod_Fe_N, Sod_Fe_C	PLN02471 super family	7.9	25,159.65	−0.21	Mitochondrial
RcMSD1	XP_024157905.1	235	5	6	78,784,596	78,786,577	Sod_Fe_N, Sod_Fe_C	PLN02471 super family	6.38	26,392.03	−0.403	Mitochondrial
RcCSD1	XP_024186633.1	152	3	8	3,432,532	3,435,148	Sod_Cu	superoxide dismutase [Cu-Zn]	5.77	15,551.98	−0.197	Cytoplasmic
RcCSD3	XP_024188883.1	159	3	7	48,229,765	48,232,866	Sod_Cu	CuZn_Superoxide_Dismutase super family	6.78	15,950.84	−0.184	Cytoplasmic Extracellular
RcCSD2	XP_024158982.1	225	5	8	60,486,324	60,494,142	Sod_Cu	CuZn_Superoxide_Dismutase super family	6.02	22,896.16	0.075	Chloroplast

proteins belonging to a similar subfamily were clustered together (Díaz-Pérez et al., 2007). However, *MnSODs* and *FeSODs* were present in one sub-groups; indicating that they possess a close relationship with a common ancestor. Moreover, when compared with all other plants, *R. chinensis* SODs showed a close evolutionary relationship to *A. thaliana* in each clade.

Analysis of conserved motifs and gene structures

To check the similarities of the SOD gene in *R. chinensis*, conserved motifs analysis, and gene structures were analyzed. Only motif 6 is conserved among all the SODs while motifs 3 and 5 are conserved in all *RcSODs* (Fig. 3B). Motifs 2,7,5,6 and 1 are present in all members of FSDs except *AtFSD2* (motifs 5 and 1 are present). All the subfamilies of MSDs have motifs 2,10,5,7,1,8 and 6. In all the subfamilies of CSDs motifs 1,5,4,3 and 6 are present except in *RcCSD3* and *OsCSD1* motif 1 is missing (Fig. 3B). Similarly, the number of exons and introns organization was also determined in *RcSOD*. The number of exons varied in seven putative *RcSOD* ranging from 6 to 9 In RcMSD1 and RcMSD2 have a lower number of exons that is 6 while RcFSD1 has 9 number exons (Fig. 3A)

Chromosomal mapping and syntenic analysis

The 7 genes of RcSOD were located on four chromosomes, *RcFSD1* and *RcMSD2* are present on chromosome 1, *RcCSD1* and *RcCSD3* on chromosome 3, *RcFSD3* on chromosome 4, and *RcMSD1* and *RcCSD2* are located on chromosome 5. We studied the syntenic blocks of SOD genes in *R. chinensis, A. thaliana*, and *G.* max to evaluate the links between SOD genes family and its gene duplications in conjunction with our earlier findings (Fig. 4). Intra-genome syntenic study revealed that four *R. chinensis* SODs (*RcMSD1, RcFSD3, RcCSD2, and RcCSD1*) were involved in segmental-duplication events. Segmental duplications were crucial in the growth of SOD genes in the genome of *R. chinensis* (Jiang et al., 2019). The segmental duplication occurrences could offer evidence in favor of functional divergences' tighter control of SOD activity under varied stress situations. Tandem duplications were also discovered in *G.* max but not in *RcSODs* (Neto et al., 2018).

Promoter analysis in RcSOD

To predict the role of *RcSODs* cis-regulatory elements, a 2 kbp upstream region of SOD genes from *R. chinensis* was taken. PlantCARE tool (" https://bioinformatics.psb.ugent.be/webtools/plantcare/html/") was used to identify the transcriptional regulatory elements in *RcSODs* (Lescot et al., 2002). A total of 11 cis-elements are identified in RcSODs (Table 2) that have a potential role in various stress responses and transcription factor (TF) binding (Kasowski et al., 2010). The cis-regulatory elements interact with transcription factors and activate the mechanism of stress tolerance. Most of the cis-acting elements possess TATA and CAAT as a core sequence in the enhancer region of the promoter (Fang et al., 1989).

Expression analysis of RcSODsand qRT-PCR

To get an insight into the functional properties of RcSODs, relevant expression patterns were calculated from RNA-seq data and presented using a heat map by fold change (Fig. 5A). The expression analysis was performed under salt stress in the leaf and root. All seven SODs genes were upregulated in roots. However, in leaves, these genes depicted either a decrease or no variation of expression under salt stress. The *RcCSD1, RcCSD3,* and *RcFSD3* were highly upregulated in roots at 4 h (Fig. 5A). Real-time qRT-PCR based expression profiling identified similar expression profiles with minor variations. In leaves, only *RcCSD1* and *RcCSD2* were upregulated. In roots, all SODs (except RcCSD2) were significantly upregulated after 4 h of salt stress (Fig. 5B).

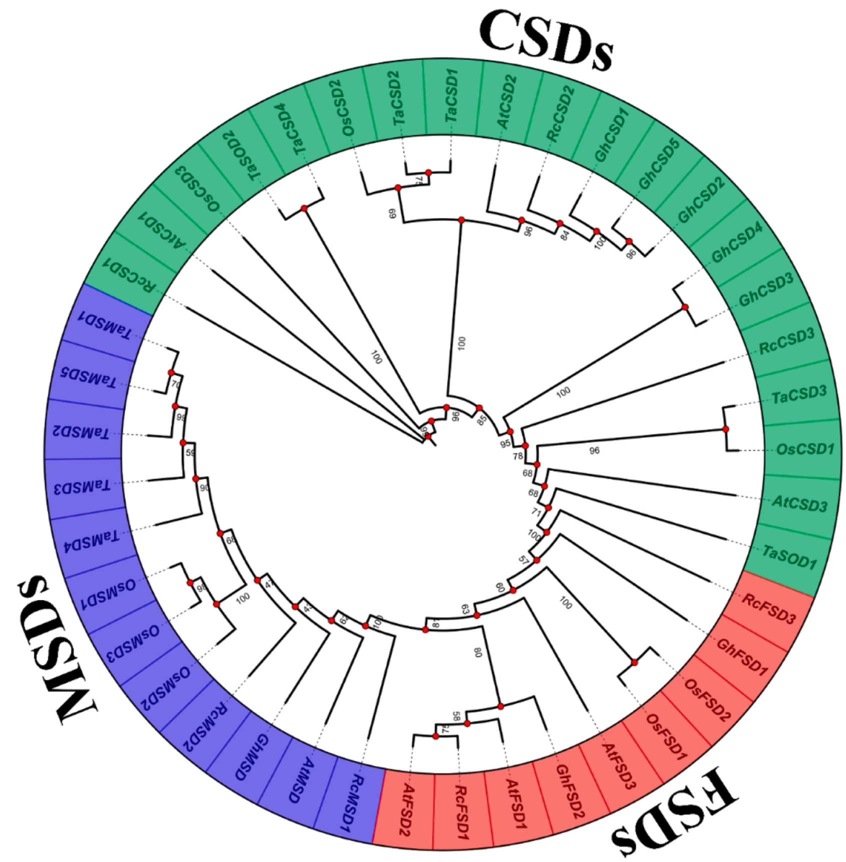

Fig. 2. Phylogenetic tree of SOD genes from *R. chinensis* with four other species. Three colors represent the three groups of *Cu/ZnSODs, MnSODs*, and *FeSODs. Cu/ZnSODs* were represented in green color, *MnSODs* were represented in red color and *FeSODs* were represented in blue color.

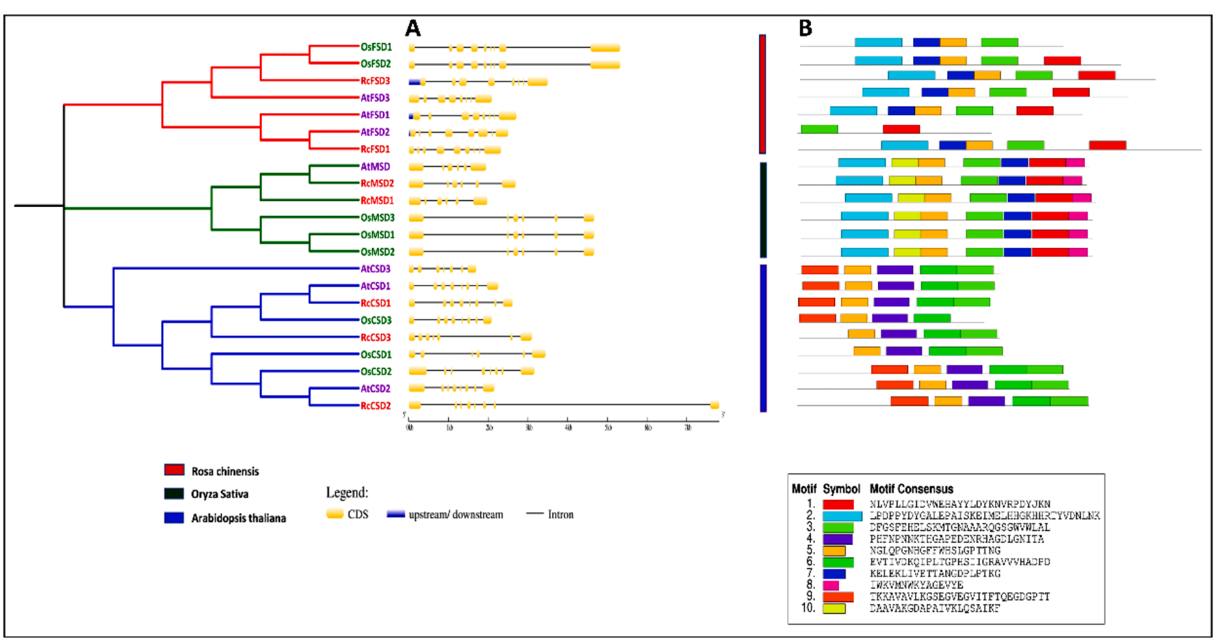

Fig. 3. *RcSOD* gene study using phylogenetic trees, gene structure, and motif analysis. It is demonstrated how *O. sativa, R. chinensis,* and *A. thaliana* evolved together. (A) The gene structure of SOD genes in *R. chinensis*; the exon is shown in a yellow box and is delineated by thin lines. UTRs are shown with blue boxes. (B) conserved motifs analysis.

3D protein prediction and docking analysis

The 3D structures of protein were predicted using I-tasser and were refined using Galaxy refine ("https://galaxy.seoklab.org/cgi-bin/submit.cgi?type=REFINE"). By adjusting the secondary structure units and rearranging sidechains, 3D models of proteins can frequently be

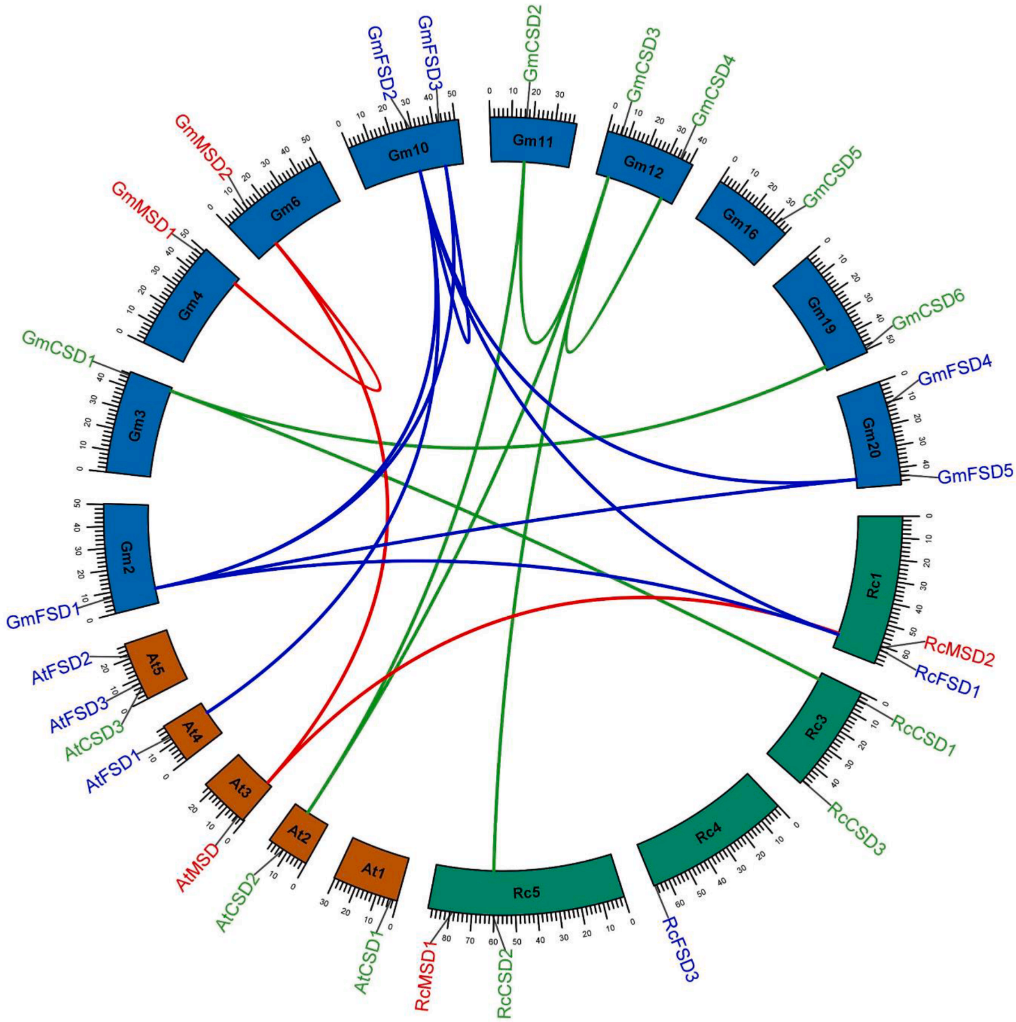

Fig. 4. Syntenic blocks of SOD genes in *R. chinensis, G.* max*,* and *A. thaliana* are represented with different colors i.e., green, blue, and red respectively. These curve lines show the gene duplication with various plants.

Table 2
Identification of *cis*-regulatory elements.

Regulatory Element	*Core sequence*	*RcCSD1*	*RcCSD2*	*RcCSD3*	*RcFSD1*	*RcFSD3*	*RcMSD1*	*RcMSD2*	*Functions*
ABRE	ACGTG	1			3	2		1	Cis-acting component contributing to the response to abscisic acid
ARE	AAACCA	1	4	5	2	3	2	6	Necessary cis-acting regulatory component for the onset of anaerobic metabolism
Box 4	ATTAAT		2		2	1	1		A conserved DNA region that is a component of the light responsiveness
CAAT-box	CAAAT	14	11	10	9	12	13	7	In the promoter and enhancer regions, a common cis-acting element
	CAAT							24	
	CCAAT	8	6	6	6	3	3	10	
CAT-box	GCCACT	1	1	1			1	2	Regulatory component that acts trans-con and controls meristem expression
CGTCA-motif	CGTCA	1	2				2	5	A cis-acting regulatory component that contributes to MeJA responsiveness
GT1-motif	GGTTAA		1	2	2	2	1		Light responsive element
MBS	CAACTG	3	2	1	1			1	Drought-inducibility is caused by the MYB binding site
P-box	CCTTTTG	1			2	2		1	Responsive to gibberellin
TATA-Box	TATACA	3			1			1	About −30 of the transcription initiations, the main promoter element
	TATAA	5	1	3	2		3		
	TATAAA	3		2			2		
	ATATAT			1	3			2	
	ATATAT			1		3	2		
TGACG-motif	TGACG	1	2				2	1	MeJA sensitivity is mediated by a cis-acting regulatory component

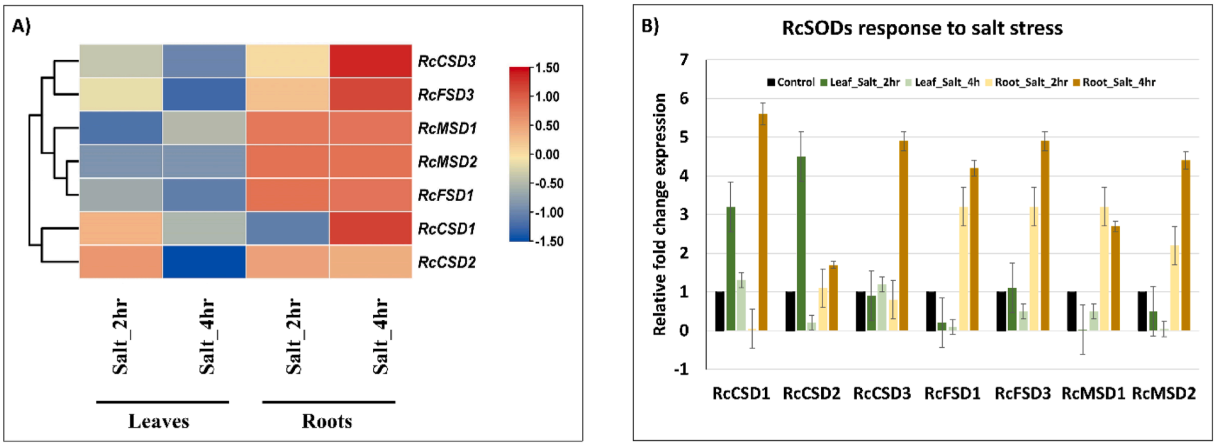

Fig. 5. Expression profile of RcSODs under salt stress. (**A**) RNA-seq-based expression profile of RcSOD. The red color represents the gene upregulation and the blue represents the down-regulation. The yellowish color represents no significant variation of expression. The scale bar represents row-wise relative FPKM fold change **(B)** Real-time qRT-PCR based expression profiling of RcSODs.

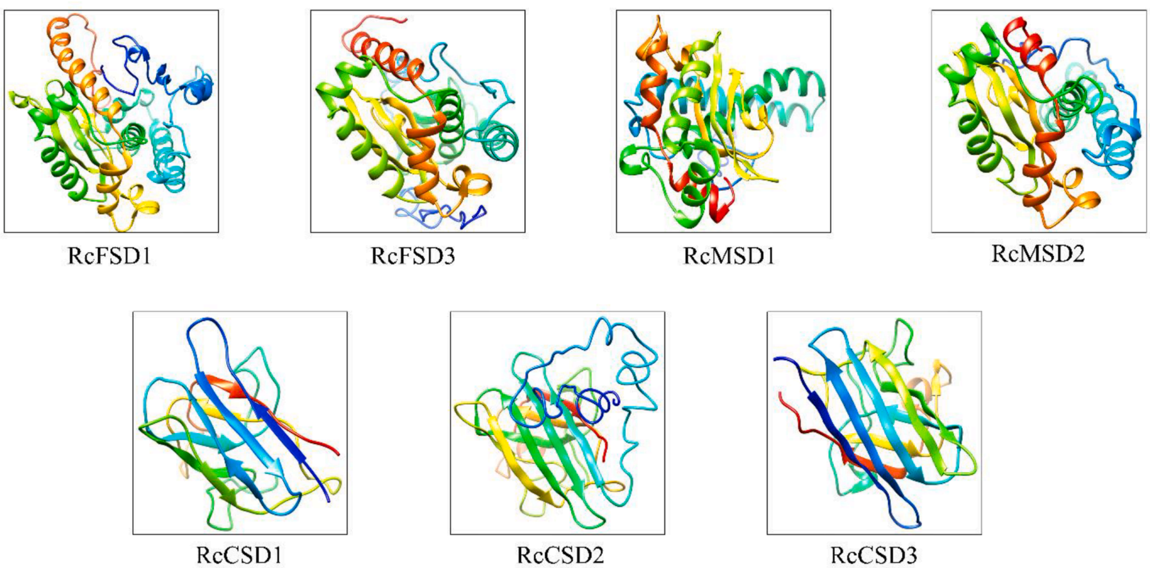

Fig. 6. Using I-TASSER, we predicted the 3D structures of RcSOD proteins. The resulting models depict the alpha helices, beta-plated sheets, and loops, each represented with distinct colors for easy visualization.

improved to get them closer to their original structure (Anand et al., 2022). The yellow and green colors represent the helices, and sheets or strands respectively. FSDs and MSDs have similar structures with the same numbers of helices and sheets and subfamilies of CSDs have a similar structure. The refined structures of the protein are shown in (Fig. 6). After refining the structures, the docking analysis was performed with the peroxyl radical.

Molecular docking analysis of RcSOD protein

Peroxyl radical was docked against the seven RcSOD protein structures. The results of molecular docking demonstrated that peroxyl radical has shown good binding capability (Ampadu et al., 2022). Docking complexes have −3.10 Kcal/Mol (RcMSD1) to −4.40 Kcal/Mol (RcCSD1) global energy and attractive VdW energy ranging from −1.71 Kcal/Mol (RcMSD1) to −2.85 Kcal/Mol (RcFSD1). Conserved residues of RcFSD1 are LYS 31, LYS53, GLN 49, GLN51 and with bond distances 2.343, 2.806, 3.059, and 2.161 Å respectively. The conserved residue of RcFSD3 is LYS40 and its bond distance is 2.249. RcMSD1 has conserved residues VAL29, PHE35, and LEU22 having bond distances of 2.897, 2.321,2.504 Å respectively. RcMSD2 has conserved residue VAL177 with bond distances of 2.391 Å. RcCSD1 has conserved residue GLN102 with bond distances of 2.303 Å. RcCSD2 has conserved residue ILE27 with bond distances of 3.240 Å. Conserved residues of RcCSD3 are VAL103, LEU88, and ASN90 with the bond distances 2.549, 2.533, and 3.181 Å respectively. (Fig. 7)

Discussion

R. chinensis is the major flowering plant globally and encounters various environmental and biological stresses, ultimately resulting in a reduction of its yield (Cheng et al., 2022). Moreover, a rapid decline in cultivatable agricultural land has compelled farmers to grow these non-cash crops on marginal lands. Hence, it is necessary to understand the biological mechanisms of stress tolerance in these plants. Under stress and normal conditions, reactive oxygen species (ROS) are produced and controlled by SODs (Mates, 2000). SODs have a role in the response towards biotic and abiotic stress and represent the first line of defense against ROS. SODs are key components of the antioxidant defense system, countering ROS under stress and normal conditions.

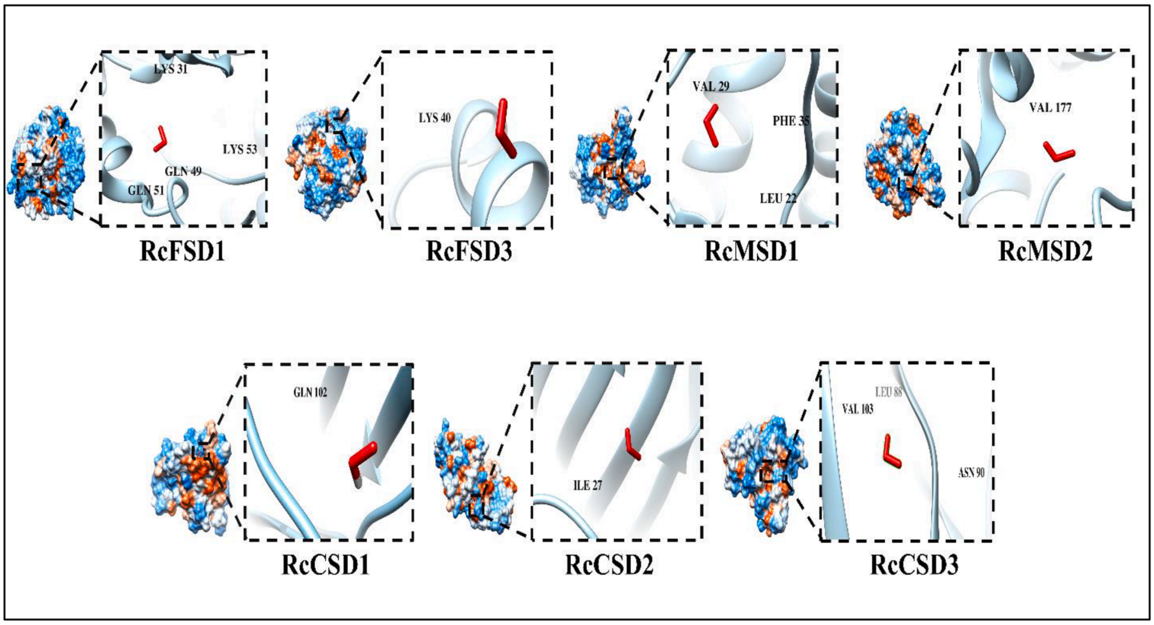

Fig. 7. The interaction between RcSOD proteins and peroxyl radicals was investigated through molecular docking. The binding sites (residues) of the proteins were visualized using Chimera software, and the peroxyl radicals were prominently displayed in red color.

Identifying and categorizing seven putative SOD genes lays the foundation for investigating stress responses (Van Raamsdonk and Hekimi, 2010). Comparative analysis reveals insights into SOD gene evolution and adaptive significance, with segmental duplications implying functional diversification and tighter regulation under stress (Gaca et al., 2019). These findings advance plant biology knowledge and offer potential for biotechnological interventions to enhance stress tolerance in crops and sustainable agriculture practices. In *R. chinensis*, the identification and characterization of SODs are not yet reported. The genome-wide identification of SOD genes is facilitated by the availability of whole genome sequence data. Pfam and Blast search (predict the well-known proteins of related families) are the basic pipelines to identify the SOD genes in *R.chinensis* (Aleem et al., 2022; Zhou et al., 2022). In the present study, we identified seven putative genes of SOD in the *R. chinensis* genome and divided them into three groups (3CSDs, 2 FSDs, and 2 MSDs) based on phylogenetic tree and domain analysis. Moreover, in previous studies SODs were reported in *Sorghum bicolor* (Huo et al., 2022), *A. thaliana* (Cui et al. 2023), *T. aestivum* (Aleem et al., 2022), tomato (G. Zhou et al., 2022), Longan (Rehman et al., 2022), cucumber (Luo et al., 2022), rice (Tounsi et al., 2022), banana (Niu et al., 2022), and *G. hirsutum* (Wang et al., 2017). The members of SOD family varied may be due to the divergence in the size of the genome in different species.

The analysis of physicochemical properties like molecular weight, pI, GRAVY, and amino acid length of SODs proteins determined in *R. chinensis* showed similar trends as predicted in earlier studies like *Saliva miltiorrhiza* (Niu et al., 2022). The subcellular localization of *RcSODs* was similar to those identified in previous studies. In plants, CSDs were mostly found in chloroplast, cytoplasm, and peroxisomes (Raja et al., 2022; Rehman et al., 2022). MSDs were associated with the mitochondrion reported in *Brassica juncea* (Niu et al., 2022). In cowpea, FSDs have been disclosed to be present in cytoplasm and chloroplast.

Phylogenetic analysis showed a close relationship between the members of Fe-SODs/Mn-SODs and Cu/Zn-SODs. Comparative phylogenetic analysis of *R. chinensis* and other four plant species (*A. thaliana, O. sativa, G. hirsutum,* and *T. aestivum)* revealed that SOD is divided into two clades with ML method based on bootstrap value. Moreover, these findings are consistent with earlier research (Huo et al., 2022; Niu et al., 2022). Motifs 3 (DFGSFEHELSKMTGNAAARQGSGWVWLAL) and 5 (NGLQPGNHGFFWHSLGPTTNG) are highly conserved in all *RcSODs*. Whereas, MSD, FSD, and CSD proteins showed conserved motifs and were reported in grapevine (Rehman et al., 2022). In *RcSODs*, the Sod_Cu domain (Pfam: PF00080) was conserved in all CSD. The C-terminal iron/manganese SOD domain (Pfam: PF02777) and N-terminal iron/manganese SOD-alpha hairpin domain (Pfam: PF00081) present in 2 MSD and 1 FSD in *RcSODs* and previously reported in S. *miltiorrhiza* (Niu et al., 2022). However, the Sod_Fe_N Superfamily domain exists in FSD 2. Gene architecture revealed that SOD genes possess a variable number of introns ranging from 5 to 8. It was reported that genes of SOD contain highly conserved patterns of introns and most chloroplastic and cytosolic SODs possess seven introns (Jiang et al., 2019).

Cis-regulatory elements are also known as molecular switches that have a potential role in the transcriptional regulation activity of genes (Wittkopp and Kalay, 2012). These elements are involved in the regulation of various biological processes in the plant body such as hormone responses, development processes, and stress responses (Yamaguchi-Shinozaki and Shinozaki, 2005). In *R. chinensis,* different stress-responsive regulatory elements were predicted and named as ARE, ABRE, Box 4, CAAT-box, CAT-box, CGTCA-motif, GT1-motif, MBS, P-box, TATA-box, and TGACG-motif. In tomatoes, transcription factor ERF binds to GCC-box to offer salt stress tolerance (Jin et al., 2022). These *cis*-regulatory elements help to understand their role in mechanisms related to biotic and abiotic stress. Syntenic analysis with *A. thaliana* and *G.* max showed segmental duplications in four *R. chinensis* SOD genes (RcMSD1, RcFSD3, RcCSD2, and RcCSD1), suggesting their critical role in gene family growth and functional diversification. This supports tighter control of SOD activity under varied stress conditions, essential for adaptive responses in plants. These findings provide insights into the evolutionary dynamics of the SOD gene family and its potential implications for stress tolerance in plants. The protein structures exhibited a conserved pattern in their loops, helices, and sheets among the same members. Specifically, RcFSD1 and RcFSD3 appeared to be highly similar. RcMSD1 and RcMSD2 also look alike and the same is the case with RcCSD1, RcCSD2, and RcCSD3. *RcCSD1* and *RcCSD2* are significantly upregulated under salt stress in roots while *RcCSD1, RcCSD3,* and *RcFSD3* genes under salt stress are highly up-regulated. After molecular docking analysis, it is observed that despite structural differences Valine from all proteins (different locations) is docked with peroxyl radical. Reactive oxygen species (ROS) that is damaging to plant growth and development is peroxyl radical (Imosemi, 2013). SOD is an antioxidant that helps to remove reactive ROS, such as peroxyl radicals.

Conclusion

In *R. chinensis* 7 SOD genes are categorized into three types (Cu/Zn-SODs, Mn-SODs, and Fe-SODs). These genes are distributed on the 7 chromosomes. In this study, SOD-specific the properties like the organization of intron-exon, subcellular localization, functional analysis, and motifs were also revealed. In the promoter region of RcSODs, stress-sensitive *cis*-regulatory elements were also discovered. The characterization of SOD family in *R. chinensis* is the first comprehensive study in understanding evolution, putative functions, and identification of SOD in *R. chinensis* under salt stress.

Authors contribution

Muhammad Umar Rafique and Nazia Nahid conceived the idea and supervised the research. The investigation was carried out by Muhammad Umar Rafique, Muhammad Waqas, Farrukh Azeem, and Muhammad Tahir ul Qamar. Muhammad Umar Rafique, and Farrukh Azeem recorded data and did formal analysis. Sajid Fiaz and KOTB A. Attia provided technical expertise and helped in writing of original draft. Roshan Zameer, Asmaa M. Abushady, and Muhammad Umar Rafique helped during the revision of the article. All authors carefully read, revised, and approved article for submission.

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Research ethics

We also took appropriate permission from the farm or field owner during specimens' collection and experimentation. We confirm that during the collection and execution of the experiment, authors have complied with the IUCN Statement on Research Involving Species at Risk of Extinction and the Convention on the Trade in Endangered Species of Wild Fauna and Flora. All methods were performed in accordance with the relevant guidelines/regulations/legislation.

Ethical approval

This article does not contain any studies with human/animals performed by any of the authors.

Consent for publication

All authors approved the final manuscript and agreed to the publication.

Declaration of Competing Interest

All authors have direct intellectual contribution to the manuscript. The authors are in complete agreement to make submission with Plant Stress. The authors declare no conflict of interest.

Data availability

Data will be made available on request.

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Supplementary materials

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